

Easy (10 MCQs)

1.

What does AAA stand for in networking?

- a) Authentication, Authorization, and Accounting
- b) Access, Administration, and Audit
- c) Apply, Allow, and Authenticate
- d) None of the above

Answer: a

Explanation: AAA is the standard framework to secure network access.

2.

OVSDB is used for:

- a) Managing Open vSwitch configuration
- b) Forwarding packets
- c) Authentication
- d) Load balancing

Answer: a

Explanation: OVSDB manages OVS switch state and configs.

3.

Group Based Policy is primarily used for:

- a) Defining policies based on endpoint groups
- b) Routing packets
- c) Encrypting data
- d) User login authentication

Answer: a

Explanation: GBP simplifies network policy by grouping endpoints.

4.

Service Function Chaining means:

- a) Steering traffic through a sequence of services
- b) Creating VLANs
- c) Encrypting packets
- d) None of the above

Answer: a

Explanation: SFC chains services like firewall, DPI, etc., in order.

5.

LISP separates:

- a) Endpoint identity and location
- b) Packets and frames
- c) IP addresses and MAC addresses
- d) None of the above

Answer: a

Explanation: LISP separates device ID from network location.

6.

Virtual Tenant Networks enable:

- a) Multiple isolated virtual networks on shared infrastructure
- b) Only single tenant usage
- c) Physical cabling changes
- d) None of the above

Answer: a

Explanation: VTNs enable multi-tenant network virtualization.

7.

Which protocol is often used to secure AAA communications?

- a) RADIUS

- b) FTP
- c) SMTP
- d) DHCP

Answer: a

Explanation: RADIUS is widely used for AAA authentication and accounting.

8.

Which component manages OVS configuration via OVSDDB?

- a) ODL controller
- b) DHCP server
- c) DNS server
- d) None of the above

Answer: a

Explanation: ODL uses OVSDDB to manage virtual switches.

9.

Application intents in ODL represent:

- a) High-level desired network outcomes
- b) Low-level device commands
- c) User login credentials
- d) None of the above

Answer: a

Explanation: Intents declare *what* to do, not *how*.

10.

In SFC, which protocol is used to track service chains?

- a) NSH (Network Service Header)
- b) HTTP
- c) TCP
- d) FTP

Answer: a

Explanation: NSH encapsulates packets for service chaining.

Medium (20 MCQs)

11.

Which ODL project supports policy management based on endpoint groups?

- a) Group Based Policy (GBP)
- b) LISP
- c) Mininet
- d) None of the above

Answer: a

Explanation: GBP is designed for policy based on groups.

12.

What does the OVSDB protocol primarily communicate?

- a) Switch configuration data
- b) Packet forwarding decisions
- c) User authentication info
- d) Network monitoring data

Answer: a

Explanation: OVSDB manages switch configurations.

13.

How does OpenDaylight enforce AAA?

- a) By integrating with external AAA servers like RADIUS

- b) By changing switch firmware
- c) By blocking all users by default
- d) None of the above

Answer: a

Explanation: ODL integrates with AAA servers for authentication.

14.

What is a primary benefit of virtual tenant networks?

- a) Tenant isolation and network slicing
- b) Increasing physical link speed
- c) Reducing packet size
- d) None of the above

Answer: a

Explanation: VTNs provide isolated virtual networks on shared infrastructure.

15.

LISP mapping service in ODL helps with:

- a) Dynamic resolution of endpoint locations
- b) Packet encryption
- c) User authorization
- d) None of the above

Answer: a

Explanation: It maps IDs to locators for routing.

16.

Which protocol encapsulates packets in SFC?

- a) NSH
- b) TCP
- c) UDP
- d) HTTP

Answer: a

Explanation: NSH is the encapsulation header for service chaining.

17.

What is the function of application intents?

- a) Translate business policies into network configurations
- b) Authenticate users
- c) Monitor network performance
- d) None of the above

Answer: a

Explanation: Intents abstract policies into network actions.

18.

Which protocol is commonly used to implement AAA?

- a) TACACS+
- b) SMTP
- c) FTP
- d) DHCP

Answer: a

Explanation: TACACS+ is another AAA protocol besides RADIUS.

19.

In ODL, what is a typical use case for OVSDB?

- a) Virtual switch management and configuration
- b) Packet filtering
- c) Routing protocols
- d) None of the above

Answer: a

Explanation: OVSDB manages OVS settings remotely.

20.

Which of these is NOT a component of AAA?

- a) Billing only (not accounting)
- b) Authentication
- c) Authorization
- d) Accounting

Answer: a

Explanation: Billing alone is not part of AAA; accounting includes usage tracking.

21.

How does ODL support Service Function Chaining?

- a) By orchestrating network service paths and managing NSH headers
- b) By encrypting packets only
- c) By managing physical cabling
- d) None of the above

Answer: a

Explanation: ODL manages SFC by steering traffic and adding NSH encapsulation.

22.

What is the role of the LISP control plane in ODL?

- a) Managing ID-to-locator mappings dynamically
- b) Encrypting network data
- c) Managing user sessions
- d) None of the above

Answer: a

Explanation: LISP control plane resolves identifiers to locations.

23.

What is the primary goal of Virtual Tenant Networks?

- a) Multi-tenant network virtualization with isolation
- b) Increasing physical port counts
- c) Encrypting wireless signals
- d) None of the above

Answer: a

Explanation: VTNs create isolated virtual networks for tenants.

24.

Which project in ODL helps define policy in terms of groups rather than individual endpoints?

- a) Group Based Policy
- b) Mininet
- c) Karaf
- d) None of the above

Answer: a

Explanation: GBP focuses on groups for policy simplification.

25.

How does ODL interact with OVS for virtualization?

- a) Via OVSDDB for switch configuration
- b) Via DHCP for IP assignment
- c) Via FTP for file transfers
- d) None of the above

Answer: a

Explanation: OVSDDB is the management protocol for OVS.

26.

What is the main advantage of Application Intents in ODL?

- a) Simplifies network programming by abstracting details
- b) Encrypts network traffic
- c) Provides user login authentication

d) None of the above

Answer: a

Explanation: Intents allow high-level network intent without low-level configs.

27.

Service Function Chaining typically involves:

a) Multiple service nodes arranged in sequence

b) Single service node only

c) Only physical devices

d) None of the above

Answer: a

Explanation: SFC chains multiple services through which traffic flows.

28.

Which protocol is NOT related to AAA?

a) FTP

b) RADIUS

c) TACACS+

d) LDAP

Answer: a

Explanation: FTP is a file transfer protocol, not AAA.

29.

Which of the following is a use case for LISP in SDN?

a) Network mobility and multi-homing

b) Packet filtering

c) Firewall configuration

d) None of the above

Answer: a

Explanation: LISP supports endpoint mobility and locator separation.

30.

How are virtual networks isolated in ODL's VTN project?

- a) Through logical separation at the controller level
- b) Using physical cables only
- c) Using separate hardware for each tenant
- d) None of the above

Answer: a

Explanation: VTN provides logical multi-tenancy.

Hard (10 MCQs)

31.

How does ODL maintain AAA state consistency across distributed deployments?

- a) Through integration with centralized AAA servers and replication
- b) By manual synchronization only
- c) By disabling AAA in clusters
- d) None of the above

Answer: a

Explanation: AAA state is maintained via centralized servers and replication.

32.

What challenges arise in OVSDb virtualization with multiple tenants?

- a) Ensuring isolation and resource allocation dynamically
- b) Encrypting traffic only
- c) User authentication

d) None of the above

Answer: a

Explanation: Virtualization requires isolating tenant traffic and resources.

33.

In Group Based Policy, what does policy enforcement rely on?

- a) Endpoint groups and associated contracts
- b) Physical switch configuration
- c) Static ACLs only
- d) None of the above

Answer: a

Explanation: GBP uses groups and contracts for dynamic policy.

34.

How does Service Function Chaining interact with SDN controllers?

- a) Controllers program forwarding rules to enforce chain order
- b) Only physical devices are configured manually
- c) SFC does not interact with SDN controllers
- d) None of the above

Answer: a

Explanation: SDN controllers like ODL program forwarding paths for SFC.

35.

LISP flow mapping in ODL is beneficial because:

- a) It allows location transparency and flexible routing
- b) It encrypts all flows
- c) It reduces packet size
- d) None of the above

Answer: a

Explanation: LISP decouples identity from location improving routing.

36.

How does ODL implement Virtual Tenant Networks internally?

- a) By maintaining tenant-specific logical routers, switches, and links in MD-SAL
- b) By physical network reconfiguration only
- c) By user authentication only
- d) None of the above

Answer: a

Explanation: ODL uses MD-SAL to model tenant virtual networks.

37.

What is a key limitation when using application intents in complex networks?

- a) Difficulty in expressing very low-level device-specific features
- b) They always require manual configuration
- c) Intents are only for wireless networks
- d) None of the above

Answer: a

Explanation: Intents abstract complexity but may not expose all device features.

38.

Which security concern must be addressed in AAA integration with ODL?

- a) Secure communication channels to AAA servers
- b) Increased packet loss
- c) Reduced bandwidth
- d) None of the above

Answer: a

Explanation: Communication to AAA servers must be encrypted and authenticated.

39.

In SFC, what is the function of the Network Service Header (NSH)?

- a) It carries metadata about the service path and packet context
- b) It replaces IP headers
- c) It encrypts packets
- d) None of the above

Answer: a

Explanation: NSH provides context for packets traveling through service chains.

40.

How can LISP improve scalability in large SDN networks?

- a) By reducing routing table size via ID/locator separation
- b) By increasing packet size
- c) By requiring manual route configuration only
- d) None of the above

Answer: a

Explanation: LISP reduces routing complexity by separating identity and location.